

Hwk4 work

April 2, 2026

1 Import packages + Setup data for the rest of the assignment

```
[1]: !pip -q install rpy2
      %load_ext rpy2.ipython
```

```
[notice] A new release of pip is
available: 24.2 -> 26.0.1
[notice] To update, run:
pip install --upgrade pip
```

```
[2]: %%R
      options(repos = c(CRAN = "https://cloud.r-project.org"))

      if (!require("pacman")) install.packages("pacman")
      pacman::p_load(tidyverse, ggplot2, dplyr, lubridate)
```

Loading required package: pacman

```
[3]: %%R
      library(readr)
      library(dplyr)
      library(tidyr)
      library(ggplot2)
      library(lubridate)
      library(stringr)
      library(scales)
      library(broom)
      library(AER)

      extract_term <- function(model, term_name, spec, model_name) {
        tidy(model) %>%
          filter(term == term_name) %>%
          mutate(spec = spec, model = model_name) %>%
          select(model, spec, estimate, std.error, statistic, p.value)
      }
```

```

final.hcris.v1996 <- read_csv("../data/HCRIS_Data_v1996.txt", show_col_types =  

  FALSE) %>%  

  mutate(  

    hvbp_payment = NA_real_,  

    hrrp_payment = NA_real_  

  )  

final.hcris.v2010 <- read_csv("../data/HCRIS_Data_v2010.txt", show_col_types =  

  FALSE)  

final.hcris <- bind_rows(final.hcris.v1996, final.hcris.v2010) %>%  

  mutate(  

    fy_end = mdy(fy_end),  

    fy_start = mdy(fy_start),  

    date_processed = mdy(date_processed),  

    date_created = mdy(date_created),  

    provider_number = str_trim(as.character(provider_number)),  

    tot_discounts = abs(tot_discounts),  

    hrrp_payment = abs(hrrp_payment),  

    fyear = year(fy_end),  

    report_date = coalesce(date_processed, date_created, fy_end)  

  ) %>%  

  filter(  

    !is.na(provider_number),  

    provider_number != "",  

    fyear >= 2009,  

    fyear <= 2019  

  )  

final.hcris.clean <- final.hcris %>%  

  arrange(provider_number, fyear, desc(report_date), desc(fy_end)) %>%  

  group_by(provider_number, fyear) %>%  

  slice(1) %>%  

  ungroup() %>%  

  mutate(  

    discount_factor = 1 - tot_discounts / tot_charges,  

    price_num = (ip_charges + icu_charges + ancillary_charges) *  

    discount_factor - tot_mcare_payment,  

    price_denom = tot_discharges - mcare_discharges,  

    price = price_num / price_denom,  

    price = if_else(is.finite(price) & price > 0 & price_denom > 0, price,  

    NA_real_),  

    net_penalty = coalesce(hrrp_payment, 0) - coalesce(hvbp_payment, 0)  

  )

```

Attaching package: 'scales'

The following object is masked from 'package:purrr':

discard

The following object is masked from 'package:readr':

col_factor

Loading required package: car

Loading required package: carData

Attaching package: 'car'

The following object is masked from 'package:dplyr':

recode

The following object is masked from 'package:purrr':

some

Loading required package: lmtest

Loading required package: zoo

Attaching package: 'zoo'

The following objects are masked from 'package:base':

as.Date, as.Date.numeric

Loading required package: sandwich

Loading required package: survival

2 Summarize the Data

2.1 Question 1

2.1.1 From 2009-2019, how many hospitals filed more than one report in the same year? Show your answer as a line graph of the number of hospitals over time.

```
[4]: %>%  
q1.plot.dat <- final.hcris %>%  
  group_by(provider_number, fyear) %>%  
  summarize(n_reports = n(), .groups = "drop") %>%  
  group_by(fyear) %>%  
  summarize(n_hospitals = sum(n_reports > 1), .groups = "drop")
```

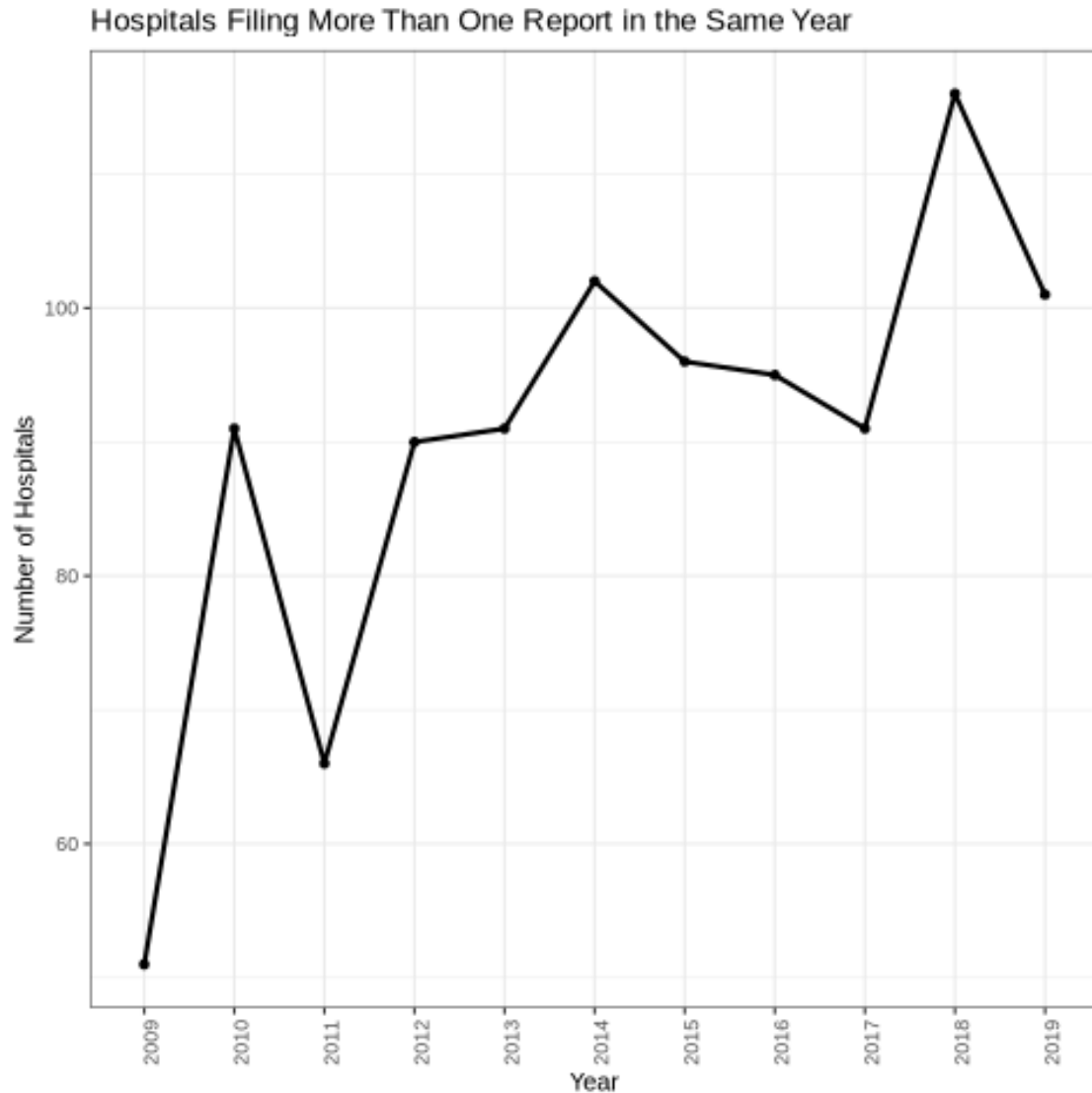
```

q1.plot.dat

q1.plot <- q1.plot.dat %>%
  ggplot(aes(x = as.factor(fyear), y = n_hospitals, group = 1)) +
  geom_line(linewidth = 1) +
  geom_point() +
  labs(
    x = "Year",
    y = "Number of Hospitals",
    title = "Hospitals Filing More Than One Report in the Same Year"
  ) +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1))

q1.plot

```



2.2 Question 2

2.2.1 After removing/combining multiple reports, how many unique hospital IDs (Medicare provider numbers) exist in the data?

```
[26]: %R
final.hcris.clean %>%
  summarize(unique_hospitals = n_distinct(provider_number))

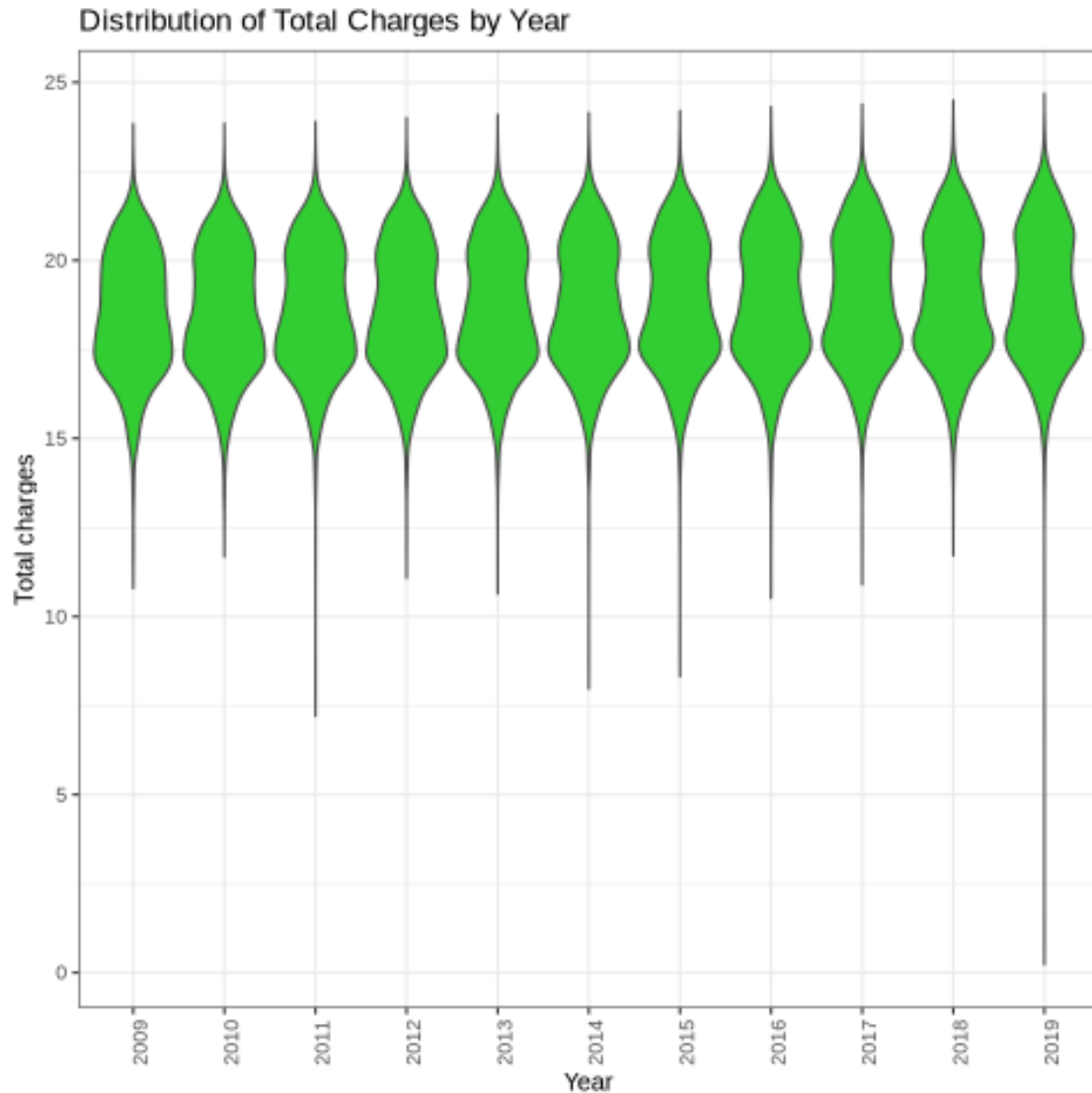
# A tibble: 1 × 1
  unique_hospitals
      <int>
1             6861
```

2.3 Question 3

2.3.1 From 2009-2019, what is the distribution of total charges (tot_charges in the data) in each year? Show your results with a “violin” plot, with charges on the y-axis and years on the x-axis.

```
[27]: %%R
charge.plot <- final.hcris.clean %>%
  filter(tot_charges > 0) %>%
  ggplot(aes(x = as.factor(fyear), y = log(tot_charges))) +
  geom_violin(fill = "limegreen", color = "gray30", trim = FALSE) +
  scale_y_continuous(labels = scales::comma) +
  labs(
    x = "Year",
    y = "Total charges",
    title = "Distribution of Total Charges by Year"
  ) +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1))

charge.plot
```



2.4 Question 4

2.4.1 From 2009-2019, what is the distribution of estimated prices in each year? Again present your results with a violin plot, and recall our formula for estimating prices from class. Be sure to do something about outliers and/or negative prices in the data.

```
[28]: %%R
price.plot.dat <- final.hcris.clean %>%
  mutate(
    discount_factor = 1 - tot_discounts / tot_charges,
    price_num = (ip_charges + icu_charges + ancillary_charges) *
    ↪discount_factor - tot_mcare_payment,
```

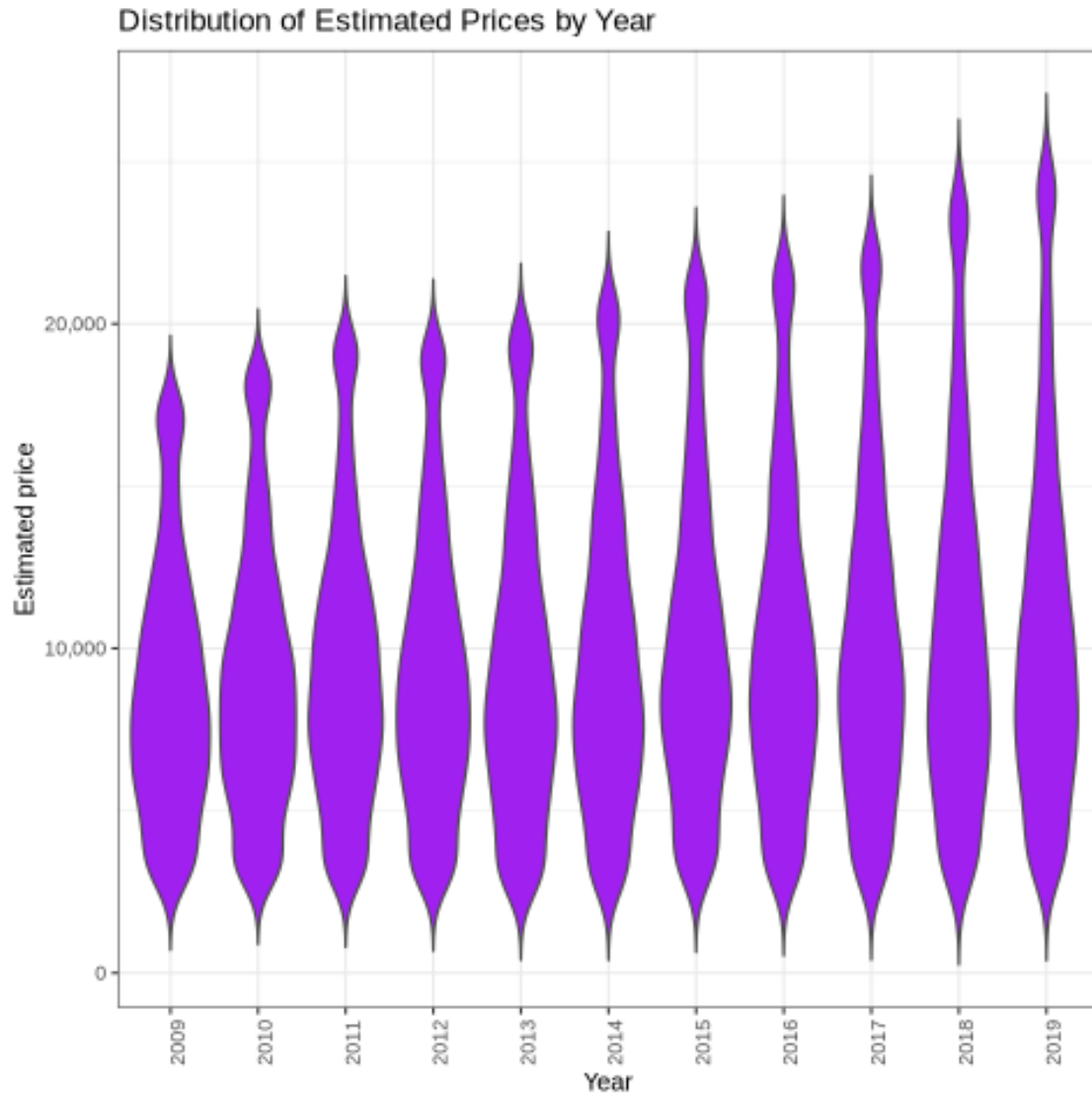
```

    price_denom = tot_discharges - mcare_discharges,
    price = price_num / price_denom
  ) %>%
  filter(price > 0, is.finite(price), price_denom > 0) %>%
  group_by(fyear) %>%
  mutate(
    p95 = quantile(price, 0.95, na.rm = TRUE),
    p05 = quantile(price, 0.05, na.rm = TRUE),
    price = pmin(pmax(price, p05), p95)
  ) %>%
  ungroup()

price.plot <- price.plot.dat %>%
  ggplot(aes(x = as.factor(fyear), y = price)) +
  geom_violin(fill = "purple", color = "gray30", trim = FALSE) +
  scale_y_continuous(labels = scales::comma) +
  labs(
    x = "Year",
    y = "Estimated price",
    title = "Distribution of Estimated Prices by Year"
  ) +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1))

price.plot

```

2.5 Question 5

2.5.1 What share of hospitals are penalized under the HRRP/VBP? Provide a graph showing the share of penalized hospitals over time, from 2012-2019.

```
[29]: %%R
share.penalty <- final.hcris.clean %>%
  mutate(net_penalty = coalesce(hrrp_payment, 0) - coalesce(hvbp_payment, 0))
  ↪ %>%
  filter(fyear %in% 2012:2019) %>%
  mutate(penalized = if_else(net_penalty > 0 & !is.na(net_penalty), 1, 0)) %>%
  group_by(fyear) %>%
```

```

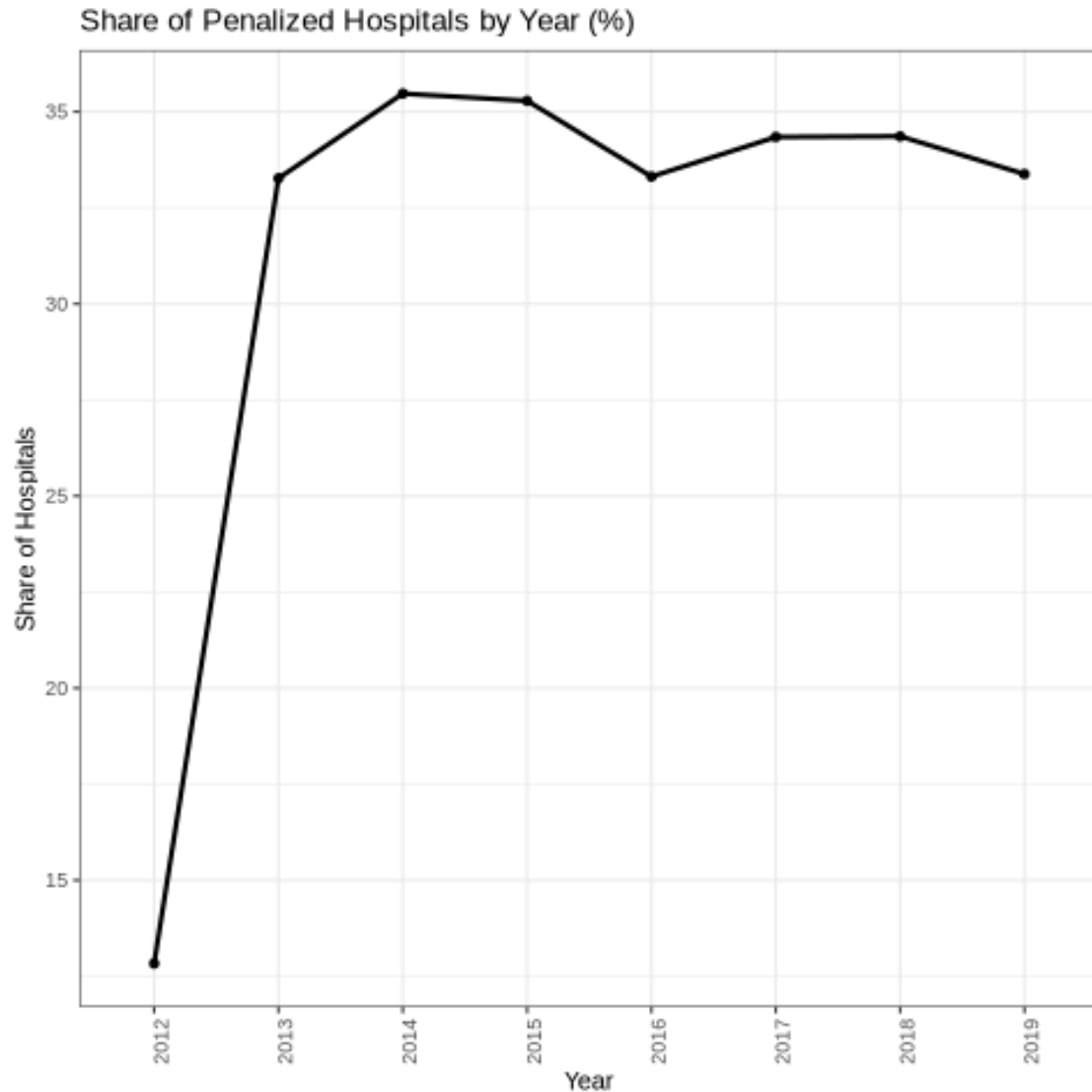
    summarize(share_penalized = mean(penalized*100, na.rm = TRUE), .groups =
↪"drop")

share.penalty

share.penalty.plot <- share.penalty %>%
  ggplot(aes(x = as.factor(fyear), y = share_penalized, group = 1)) +
  geom_line(linewidth = 1) +
  geom_point() +
  labs(
    x = "Year",
    y = "Share of Hospitals",
    title = "Share of Penalized Hospitals by Year (%)"
  ) +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1))

share.penalty.plot

```



3 Estimate ATEs

```
[31]: %>%R
### We are prepping the IV data for the last few questions
pre_period <- final.hcris.clean %>%
  filter(fyear >= 2009, fyear <= 2011) %>%
  group_by(provider_number) %>%
  summarise(
    avg_mcare_0911 = mean(mcare_discharges, na.rm = TRUE),
    avg_beds_0911 = mean(beds, na.rm = TRUE),
    avg_mcaid_0911 = mean(mcaid_discharges, na.rm = TRUE),
    .groups = "drop"
```

```

)

wide_panel <- final.hcris.clean %>%
  filter(fyear %in% c(2011, 2012, 2014)) %>%
  select(provider_number, fyear, price, net_penalty) %>%
  pivot_wider(
    names_from = fyear,
    values_from = c(price, net_penalty),
    names_sep = "_"
  )

iv_df <- wide_panel %>%
  left_join(pre_period, by = "provider_number") %>%
  transmute(
    provider_number,
    price_change = price_2014 - price_2011,
    penalty_2012 = net_penalty_2012,
    avg_mcare_0911,
    avg_beds_0911,
    avg_mcaid_0911
  ) %>%
  filter(if_all(c(price_change, penalty_2012, avg_mcare_0911), ~ !is.na(.) & is.
    ↪finite(.)))

```

3.1 Question 6

3.1.1 Provide a summary of OLS estimates of the effect of net penalties on price changes. Present your results in a table with three different specifications: 1) a “baseline” specification using only net penalty as a covariate; 2) “baseline” specification plus the pre-penalty (2009-2011) mean bed size; 3) “baseline” specification plus bed size plus pre-penalty (2009-2011) average Medicaid discharges.

```

[32] : %>%R
ols1 <- lm(price_change ~ penalty_2012, data = iv_df)
ols2 <- lm(price_change ~ penalty_2012 + avg_beds_0911, data = iv_df)
ols3 <- lm(price_change ~ penalty_2012 + avg_beds_0911 + avg_mcaid_0911, data =
    ↪iv_df)

modelsummary(
  list(
    "OLS 1" = ols1,
    "OLS 2" = ols2,
    "OLS 3" = ols3
  ),
  coef_map = c(
    "(Intercept)" = "Intercept",

```

```

    "penalty_2012" = "Net Penalty",
    "avg_beds_0911" = "Avg Beds (Pre-2012)",
    "avg_mcaid_0911" = "Avg Medicaid Discharges (Pre-2012)"
  ),
  gof_map = c("nobs", "r.squared"),
  output = "markdown"
)

```

	OLS 1	OLS 2	OLS 3
Intercept	336.811 (523.450)	299.675 (529.565)	557.145 (466.835)
Net Penalty	0.005 (0.013)	0.005 (0.013)	0.002 (0.009)
Avg Beds (Pre-2012)		0.154 (0.331)	0.115 (0.238)
Avg Medicaid Discharges (Pre-2012)			0.083 (0.174)
Num.Obs.	2493	2493	2482
R2	0.000	0.000	0.000

3.2 Question 7

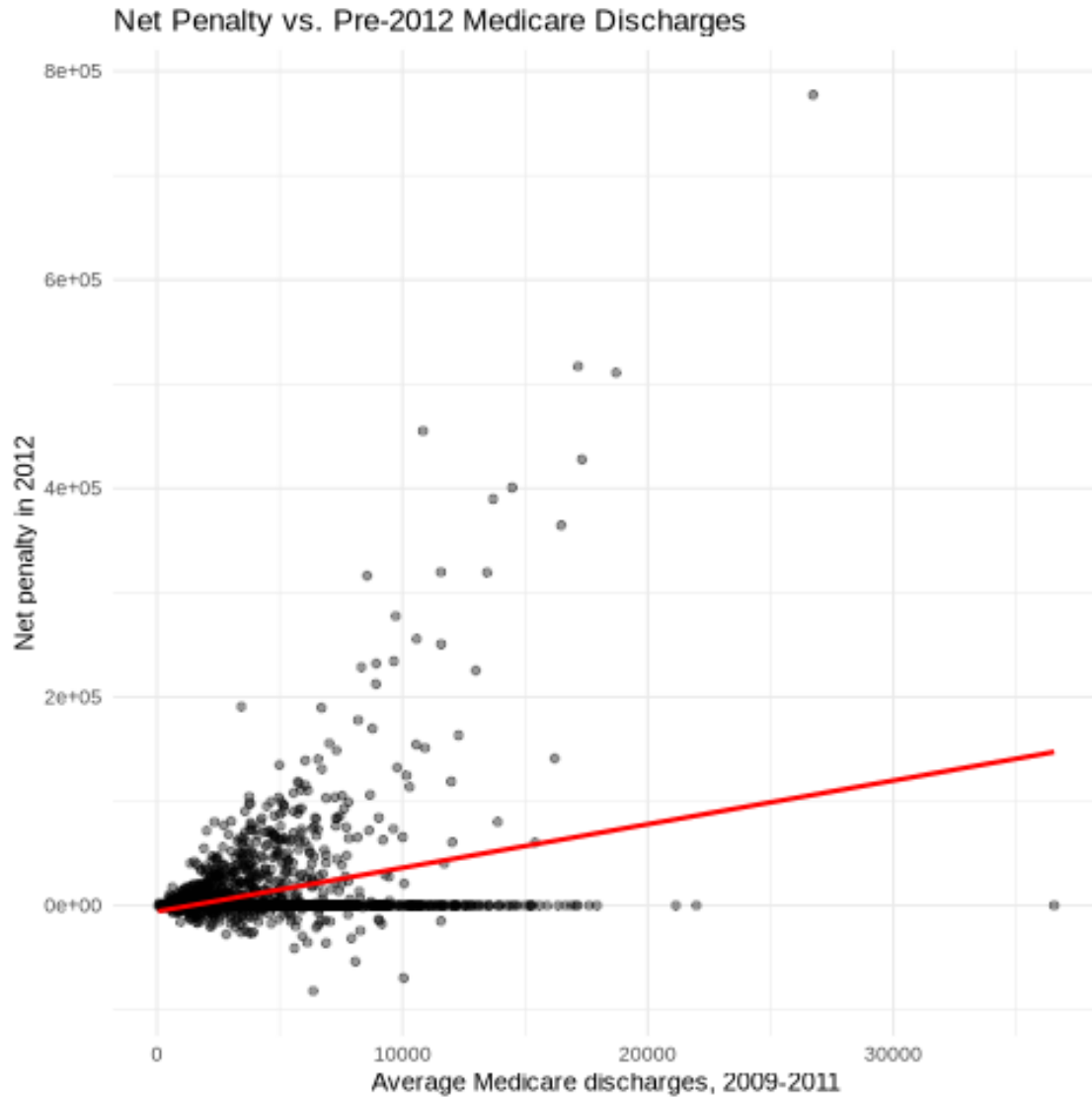
3.2.1 Provide a scatterplot of net penalty against pre-2012 Medicare discharges.

```

[33]: %>%R
ggplot(iv_df, aes(x = avg_mcare_0911, y = penalty_2012)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  labs(
    x = "Average Medicare discharges, 2009-2011",
    y = "Net penalty in 2012",
    title = "Net Penalty vs. Pre-2012 Medicare Discharges"
  ) +
  theme_minimal()

```

`geom\_smooth()` using formula = 'y ~ x'



3.3 Question 8

3.3.1 Provide a summary of the first stage and reduced-form results using pre-penalty Medicare discharges as an instrument for net penalties. Present your results in a table with three different specifications as in Question 6.

```
[34]: %>%R
library(modelsummary)

fs1 <- lm(penalty_2012 ~ avg_mcare_0911, data = iv_df)
fs2 <- lm(penalty_2012 ~ avg_mcare_0911 + avg_beds_0911, data = iv_df)
fs3 <- lm(penalty_2012 ~ avg_mcare_0911 + avg_beds_0911 + avg_mcaid_0911, data_
  ↪ iv_df)
```

```

rf1 <- lm(price_change ~ avg_mcare_0911, data = iv_df)
rf2 <- lm(price_change ~ avg_mcare_0911 + avg_beds_0911, data = iv_df)
rf3 <- lm(price_change ~ avg_mcare_0911 + avg_beds_0911 + avg_mcaid_0911, data = iv_df)

modelsummary(
  list(
    "FS 1" = fs1,
    "FS 2" = fs2,
    "FS 3" = fs3,
    "RF 1" = rf1,
    "RF 2" = rf2,
    "RF 3" = rf3
  ),
  keep = c("avg_mcare_0911"),
  coef_map = c("avg_mcare_0911" = "Pre-2012 Medicare Discharges"),
  gof_map = c("nobs", "r.squared"),
  output = "markdown"
)

```

	FS 1	FS 2	FS 3	RF 1	RF 2
Pre-2012 Medicare Discharges	4.184	4.186	4.622	0.375	0.370
0.294	(0.235)	(0.237)	(0.273)	(0.162)	(0.163)
Num.Obs.	2493	2493	2482	2493	2493
R2	0.113	0.113	0.116	0.002	0.002

3.4 Question 9

3.4.1 Provide a summary of IV estimates of the effect of net penalties on price changes. Again present your results in a table with the three different specifications as in Questions 6 and 8.

```

[35]: %>%R
iv1 <- ivreg(price_change ~ penalty_2012 | avg_mcare_0911, data = iv_df)
iv2 <- ivreg(price_change ~ penalty_2012 + avg_beds_0911 | avg_mcare_0911 + avg_beds_0911, data = iv_df)
iv3 <- ivreg(price_change ~ penalty_2012 + avg_beds_0911 + avg_mcaid_0911 | avg_mcare_0911 + avg_beds_0911 + avg_mcaid_0911, data = iv_df)

```

```

modelsummary(
  list(
    "IV 1" = iv1,
    "IV 2" = iv2,
    "IV 3" = iv3
  ),
  coef_map = c(
    "(Intercept)" = "Intercept",
    "penalty_2012" = "Net Penalty",
    "avg_beds_0911" = "Avg Beds (Pre-2012)",
    "avg_mcaid_0911" = "Avg Medicaid Discharges (Pre-2012)"
  ),
  gof_map = c("nobs", "r.squared"),
  output = "markdown"
)

```

	IV 1	IV 2	IV 3
Intercept	-453.074 (630.296)	-464.298 (632.078)	198.894 (498.206)
Net Penalty	0.090 (0.039)	0.089 (0.039)	0.064 (0.029)
Avg Beds (Pre-2012)		0.082 (0.336)	0.075 (0.241)
Avg Medicaid Discharges (Pre-2012)			-0.044 (0.185)
Num.Obs.	2493	2493	2482
R2	-0.017	-0.016	-0.017

3.5 Question 10

3.5.1 Briefly explain the “Local” ATE in the context of your estimates. How might a local effect differ from an overall ATE in this setting?

The IV estimate effectively represents the Local Average Treatment Effect. In this setting, the penalties assigned to hospitals have been impacted by their Medicare discharges before 2012. Our local ATE is informed by these values, and in that capacity, differs from our overall ATE.

In question 8, for instance, the first stage is strong, but the reduced-form and IV estimates are less statistically significant; the data do not show a strong effect for this group. This local effect could differ from the overall ATE because hospitals with higher Medicare exposure may be different from other hospitals in several factors, like their size, payer mix, and market power when it comes to price adjustment. An example in the context of these factors is a teaching hospital; these hospitals often

deal with rarer and more fatal conditions that require readmission. This could explain why some simple OLS models are biased or skewed.